

Logged in as **doctor1**

Role: **Doctor**

Log out



Obesity Level Predictor

The predictions are based on specific training data and can be incorrect.The app is aimed to help clinicians and assist them, not replace them.Regardless of the outcome, it is advised to consult to a doctor.

Patient Profile

Demographics & Habits

Gender

Male

Family history of overweight

no

Frequent high-caloric food (FAVC)

yes

Eating between meals (CAEC)

no

Smoker

yes

Monitors calorie intake (SCC)

yes

Alcohol consumption (CALC)

Frequently

Transportation used (MTRANS)

Automobile

Measurements & Frequencies

Age (years)

20.00

Height (m)

1.54

Weight (kg)

85.00

Vegetable consumption frequency (FCVC)

1.00

Number of main meals (NCP)

2.00

Daily water intake (CH2O)

1.00

Physical activity frequency (FAF)

2.00

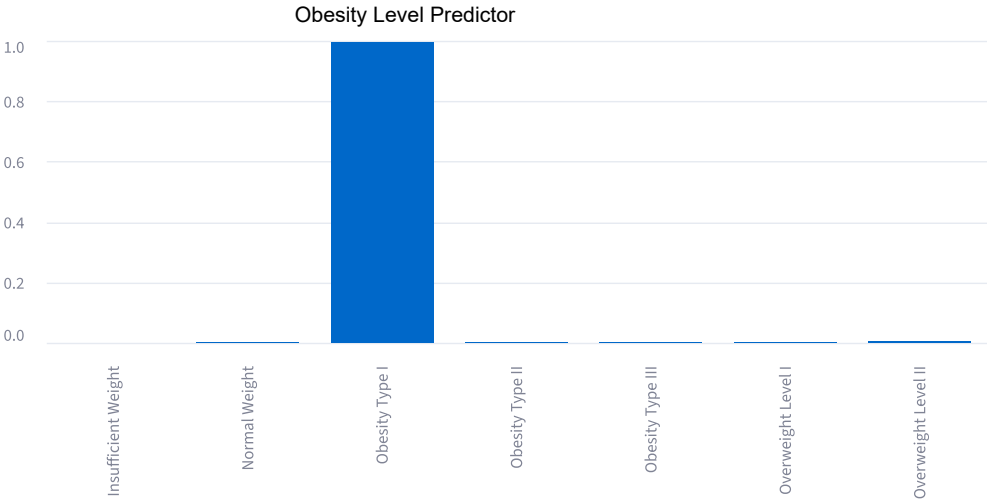
Time using technology devices (TUE)

1.00

Predict

Prediction

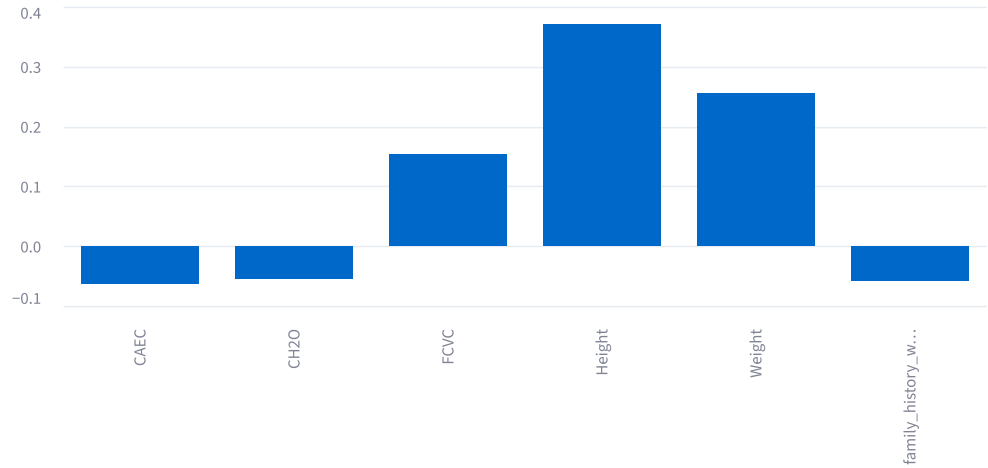
Predicted class: **Obesity Type I**



Class	Probability
Obesity Type I	99.47%
Overweight Level II	0.43%
Overweight Level I	0.06%
Obesity Type II	0.03%
Normal Weight	0.00%
Obesity Type III	0.00%
Insufficient Weight	0.00%

Why this prediction?

Top features pushing the model toward **Obesity Type I**, estimated with SHAP against a background sample from the train+val pool.



Feature	Current Value	Contribution	Direction
Height	1.540000	+0.371	toward Obesity Type I
Weight	85.000000	+0.255	toward Obesity Type I
FCVC	1.000000	+0.154	toward Obesity Type I
CAEC	no	-0.063	away from Obesity Type I
family_history_with_c	no	-0.057	away from Obesity Type I
CH2O	1.000000	-0.054	away from Obesity Type I

▼ Show submitted patient profile

Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC	MTRANS	Age
Male	no	yes	no	yes	yes	Frequen	Automobile	20

Similar Patient Profiles (top 5)

Nearest neighbors in the model's learned latent space, from the train+val pool.

latent_distance	Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC
3.052	Female	yes	yes	Sometimes	yes	no	Frequently
3.589	Female	yes	yes	Sometimes	no	no	no
3.918	Female	yes	yes	Sometimes	no	no	no
3.958	Male	yes	yes	Always	no	no	Sometimes
4.018	Female	yes	yes	Sometimes	no	no	no

Healthier Prototype

Predicted class **Obesity Type I** → one step healthier: **Overweight Level II**

Candidates with Gender=Male, Age in [18.0, 22.0], class=Overweight_Level_II: 43

Top 7 nearest 'Overweight Level II' neighbors (same gender, age ±2):

feature_distance	Gender	family_history_with_overweight	FAVC	CAEC	SMOKE	SCC	CALC
4.651	Male	no	yes	Sometimes	no	no	Sometime
4.701	Male	yes	yes	Sometimes	no	no	no
4.722	Male	yes	yes	Sometimes	no	no	no
4.725	Male	yes	yes	Sometimes	no	no	Sometime
4.751	Male	yes	yes	Sometimes	no	no	no
4.758	Male	yes	yes	Sometimes	no	no	no
4.761	Male	yes	yes	Sometimes	no	no	no

Recommended Changes

Excluded non-actionable features: Age, Height, Gender, family_history_with_overweight. Ranked by how much each feature would need to shift to look like a healthier profile.

Obesity Level Predictor

feature	current_value	prototype_value	shift_magnitude
FCVC	1.000000	2.030000	1.931
CH2O	1.000000	1.970000	1.591
MTRANS	Automobile	Public_Transportation (100% of neighbors)	1.414
CAEC	no	Sometimes (100% of neighbors)	1.414
SCC	yes	no (100% of neighbors)	1.414
SMOKE	yes	no (100% of neighbors)	1.414
CALC	Frequently	no (71% of neighbors)	1.262
NCP	2.000000	2.320000	0.409